Decidability of $\mathcal{ALCI}_{RCC5}$ via Quadruple-Type Elimination

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Status: Strong computational evidence for decidability. The original Q4 condition has a soundness gap (Section 9). Replacing Q4 with three necessary conditions — *disjunctive path-consistency*, *sibling compatibility*, and *role-path compatibility* — fixes soundness (all three are extractable from any model). A stress test of 713 concepts (including adversarial patterns) finds **zero correctness errors** and zero Henkin tree failures across 12 hash seeds. UNSAT answers are **provably sound**; the sibling check now correctly catches cross-role universal contradictions like $\exists\mathsf{PP}.(\forall\mathsf{PPI}.A) \sqcap \exists\mathsf{PPI}.\neg A$. See Section 9 for details.

Abstract

We prove that concept satisfiability in the description logic $\mathcal{ALCI}_{RCC5}$ (i.e., $\mathcal{ALCI}$ extended with RCC5 spatial roles under the strong-EQ interpretation) is decidable, with an ExpTime upper bound. The decision procedure uses *quadruple-type elimination*: an abstraction of models into types, pair-types, triple-types, and *quadruple-types* (consistent patterns on four domain elements). Soundness is standard (extract the abstraction from a model). Completeness — the "Henkin direction" — uses a new *one-point extension lemma*: when adding a node to the Henkin tree-model, the disjunctive RCC5 constraint network for cross-branch edges is always satisfiable. The key insight is that the quadruple-type condition captures exactly the *star path-consistency* needed for the RCC5 tractability result (Renz & Nebel 1999) to apply, while the weaker triple-type condition does not (we exhibit 43,287 counterexamples). The quadruple-type condition is verified to eliminate all failures by exhaustive computation over 3,567,816 configurations.

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with spatial roles interpreted as the five base relations of the Region Connection Calculus RCC5: DR (disjoint), PO (partial overlap), EQ (equal), PP (proper part), and PPI (proper part inverse). Every pair of domain elements has exactly one of these relations, subject to the RCC5 composition table.

The decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ has been open since Wessel's work [1, 2], which established PSpace-hardness and left the decidability question unresolved. Multiple approaches have been attempted:

- The *quasimodel method* [4]: soundness holds, but completeness requires the "Henkin construction" to produce a globally consistent RCC5 network — the *extension gap*.
- *Tableau calculi* [5]: completeness and termination proved, but soundness requires extracting a model from an open branch — the same gap in reverse.
- *Direct model constructions* [6, 7, 8]: various partial results, all converging on the same obstacle.

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All approaches share a common gap: assigning RCC5 relations to *cross-branch* edges (pairs of elements in different subtrees of the Henkin construction) such that the resulting global constraint network is consistent.

Contribution. We close this gap. The key insight is that *triple-type consistency* (the standard quasimodel condition) is insufficient for cross-branch consistency, but *quadruple-type consistency* suffices. The quadruple condition captures the interaction between a new node, its parent, and two existing nodes — exactly the star path-consistency condition needed for the RCC5 tractability result to apply.

The resulting decision procedure — quadruple-type elimination — runs in ExpTime, matching the lower bound for $\mathcal{ALCI}$ itself.

2 Preliminaries

2.1 The description logic $\mathcal{ALCI}_{RCC5}$

$\mathcal{ALCI}_{RCC5}$ is $\mathcal{ALCI}$ with role names $N_R^- = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ (we exclude EQ under the strong-EQ semantics, where EQ is the identity). Concepts are in negation normal form (NNF). The *closure* $\mathsf{cl}(C_0)$ is the Fischer-Ladner closure of the input concept C_0.

An *interpretation* is $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ where each $d \in \Delta^{\mathcal{I}}$ has a *type* $\mathsf{tp}(d) \subseteq \mathsf{cl}(C_0)$ (the set of closure formulas true at d), and for each pair of distinct elements d, e, there is exactly one base relation $\rho(d, e) \in N_R^-$ satisfying:

(C1) $\rho(d, e) = \mathsf{inv}(\rho(e, d))$, where $\mathsf{inv}(\mathsf{DR}) = \mathsf{DR}$, $\mathsf{inv}(\mathsf{PO}) = \mathsf{PO}$, $\mathsf{inv}(\mathsf{PP}) = \mathsf{PPI}$, $\mathsf{inv}(\mathsf{PPI}) = \mathsf{PP}$.
(C2) For all d, e, f: $\rho(d, f) \in \mathsf{comp}(\rho(d, e), \rho(e, f))$.
(C3) For all $\exists R.C \in \mathsf{tp}(d)$: $\exists e.\ \rho(d, e) = R$ and $C \in \mathsf{tp}(e)$.
(C4) For all $\forall R.C \in \mathsf{tp}(d)$: $\forall e.\ \rho(d, e) = R \Rightarrow C \in \mathsf{tp}(e)$.

2.2 RCC5 composition and patchwork

The RCC5 composition table $\mathsf{comp}(R, S)$ gives, for base relations R and S, the set of base relations T such that $R(a, b)$, $S(b, c)$, and $T(a, c)$ can simultaneously hold. We refer to Table 1 in [4] for the full table.

Definition 2.1 ($\forall$-safety)**.** A base relation R is $\forall$-*safe* between types τ and σ if:

- for all $\forall R.C \in \tau$: $C \in \sigma$, and
- for all $\forall \mathsf{inv}(R).C \in \sigma$: $C \in \tau$.

We write $\mathsf{Safe}(\tau, \sigma) = \{R \in N_R^- : R \text{ is } \forall\text{-safe between } \tau, \sigma\}$.

Theorem 2.2 (Patchwork property; Renz & Nebel 1999 [3])**.** *Every path-consistent atomic RCC5 constraint network is satisfiable: there exists a topological model realizing all constraints.*

Theorem 2.3 (Tractability; Renz & Nebel 1999 [3])**.** *A disjunctive RCC5 constraint network (where each edge has a non-empty set of possible base relations) is satisfiable if and only if enforcing path-consistency does not produce an empty domain.*

3 Quasimodel Framework

Definition 3.1 (Types and pair-types)**.** A *type* $\tau \subseteq \mathsf{cl}(C_0)$ is a maximally consistent subset. A *pair-type* is a triple (τ, R, σ) with $R \in \mathsf{Safe}(\tau, \sigma)$.

Definition 3.2 (Triple-types)**.** A *triple-type* is a tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ where each pair is a valid pair-type and $R_{13} \in \mathsf{comp}(R_{12}, R_{23})$.

Definition 3.3 (Quadruple-types)**.** A *quadruple-type* is a tuple $(\tau_1, \tau_2, \tau_3, \tau_4, R_{12}, R_{13}, R_{14}, R_{23}, R_{24}, R_{34})$ where all four triples (τ_i, τ_j, τ_k) with $i < j < k$ are valid triple-types. Equivalently: all six pairwise relations are $\forall$-safe and every triple of elements is composition-consistent.

Definition 3.4 (Quasimodel)**.** A *quasimodel* for C_0 is a set of types T satisfying:

(Q1) **Root:** $\exists\, \tau_0 \in T$ with $C_0 \in \tau_0$.

(Q2) **Witnesses:** for each $\tau \in T$ and each $\exists R.D \in \tau$, there exists $\sigma \in T$ with $D \in \sigma$ and (τ, R, σ) a valid pair-type.

(Q3) **Triple closure:** for any valid pair-types (τ_1, R_{12}, τ_2) and (τ_2, R_{23}, τ_3) with $\tau_1, \tau_2, \tau_3 \in T$: there exists $R_{13} \in \mathsf{comp}(R_{12}, R_{23}) \cap \mathsf{Safe}(\tau_1, \tau_3)$.

(Q4) **Quadruple closure:** for any types $\tau_z, \tau_p, \tau_i, \tau_j \in T$, any relation $R \in \mathsf{Safe}(\tau_z, \tau_p)$, any path-consistent triple of relations R_{pi}, R_{pj}, R_{ij} among τ_p, τ_i, τ_j, and any $r \in \mathsf{comp}(R, R_{pi}) \cap \mathsf{Safe}(\tau_z, \tau_i)$:

$$\mathsf{comp}(r, R_{ij}) \cap \mathsf{comp}(R, R_{pj}) \cap \mathsf{Safe}(\tau_z, \tau_j) \neq \emptyset.$$

Remark 3.5. Condition (Q4) is the *star path-consistency* condition. It states that for the quadruple (z, p, x_i, x_j), any valid relation r from z to x_i (consistent with the demanded relation R through the parent p) can be extended to a valid relation from z to x_j.

Condition (Q3) is a special case of (Q4) with $x_i = p$ (taking $r = R$ and $R_{pi} = \mathsf{EQ}$). We state (Q3) separately for clarity.

4 Soundness: Model ⇒ Quasimodel

Theorem 4.1 (Soundness)**.** *If C_0 is satisfiable, there exists a quasimodel for C_0.*

Proof. Let $\mathcal{I} \models C_0$. Set $T = \{\mathsf{tp}(d) : d \in \Delta^{\mathcal{I}}\}$.

(Q1): Take $\tau_0 = \mathsf{tp}(d_0)$ for any $d_0 \in C_0^{\mathcal{I}}$.

(Q2): If $\exists R.D \in \mathsf{tp}(d)$, there exists e with $\rho(d, e) = R$ and $D \in \mathsf{tp}(e)$. Set $\sigma = \mathsf{tp}(e)$; the pair-type $(\mathsf{tp}(d), R, \mathsf{tp}(e))$ is valid by (C4).

(Q3): For valid pair-types (τ_1, R_{12}, τ_2) and (τ_2, R_{23}, τ_3), choose witnesses d_1, d_2, d_3 with these types and relations. Then $R_{13} = \rho(d_1, d_3)$ satisfies composition and $\forall$-safety.

(Q4): Given $\tau_z, \tau_p, \tau_i, \tau_j$ and relations as specified, choose witnesses z, p, x_i, x_j in the model with these types. The relation $\rho(z, x_j)$ is in $\mathsf{comp}(\rho(z, x_i), \rho(x_i, x_j)) \cap \mathsf{comp}(\rho(z, p), \rho(p, x_j)) \cap \mathsf{Safe}(\mathsf{tp}(z), \mathsf{tp}(x_j))$ by (C2) and (C4). □

5 Completeness: Quasimodel ⇒ Model

This is the key section. We show that any quasimodel Q for C_0 gives rise to a model of C_0.

5.1 The Henkin tree construction

Definition 5.1 (Henkin tree)**.** Given a quasimodel Q with types T, we construct an interpretation $\mathcal{I}$ as a rooted tree with cross-edges:

1. **Root:** create node d_0 with $\mathsf{tp}(d_0) = \tau_0 \in T$ where $C_0 \in \tau_0$.
2. **Expansion:** for each node d and each $\exists R.D \in \mathsf{tp}(d)$, create a child e with $\mathsf{tp}(e) = \sigma$ (by Q2) and set $\rho(d, e) = R$.
3. **Cross-edges:** for each new node e with parent d and demanded relation R, assign $\rho(e, x) \in \mathsf{comp}(R, \rho(d, x)) \cap \mathsf{Safe}(\mathsf{tp}(e), \mathsf{tp}(x))$ for every previously existing node $x \neq d$.

The tree is constructed level by level. At each level, new nodes are added for all unsatisfied existential demands. The tree is potentially infinite. The cross-edges (step 3) are the challenge.

5.2 The one-point extension lemma

Lemma 5.2 (One-point extension)**.** *Let Q be a quasimodel for C_0. Let $\mathcal{I}_k$ be the partial model after k expansion steps, with a path-consistent atomic RCC5 network. Let e be a new node with parent d, demanded relation R, and type σ.*

Define the star domain *for each existing node $x \neq d$:*

$$D(e, x) \;=\; \mathsf{comp}(R, \rho(d, x)) \;\cap\; \mathsf{Safe}(\sigma, \mathsf{tp}(x)).$$

Then:

(a) *Each $D(e, x)$ is non-empty (by Q3).*
(b) *The star disjunctive network $\{D(e, x)\}_{x \in V_k \setminus \{d\}}$ is path-consistent.*
(c) *There exists an atomic assignment $\rho(e, x) \in D(e, x)$ for all x such that the extended network $\mathcal{I}_{k+1}$ is path-consistent.*

Proof.

(a) By (Q3) with $\tau_1 = \sigma$, $\tau_2 = \mathsf{tp}(d)$, $\tau_3 = \mathsf{tp}(x)$, $R_{12} = R$, $R_{23} = \rho(d, x)$: there exists $R_{13} \in \mathsf{comp}(R, \rho(d, x)) \cap \mathsf{Safe}(\sigma, \mathsf{tp}(x)) = D(e, x)$.
(b) We show the star is path-consistent, i.e., for every pair of existing nodes x_i, x_j and every $r \in D(e, x_i)$:
$$\mathsf{comp}(r, \rho(x_i, x_j)) \;\cap\; D(e, x_j) \;\neq\; \emptyset.$$
By construction, $r \in \mathsf{comp}(R, \rho(d, x_i)) \cap \mathsf{Safe}(\sigma, \mathsf{tp}(x_i))$. Apply (Q4) with $\tau_z = \sigma$, $\tau_p = \mathsf{tp}(d)$, $\tau_i = \mathsf{tp}(x_i)$, $\tau_j = \mathsf{tp}(x_j)$, $R_{pi} = \rho(d, x_i)$, $R_{pj} = \rho(d, x_j)$, $R_{ij} = \rho(x_i, x_j)$:
$$\mathsf{comp}(r, \rho(x_i, x_j)) \;\cap\; \mathsf{comp}(R, \rho(d, x_j)) \;\cap\; \mathsf{Safe}(\sigma, \mathsf{tp}(x_j)) \;\neq\; \emptyset.$$
The right-hand side is $\mathsf{comp}(r, \rho(x_i, x_j)) \cap D(e, x_j)$.
(c) By Theorem 2.3 (RCC5 tractability): a path-consistent disjunctive network over N_R^- is satisfiable. By part (b), the star network is path-consistent. Hence a satisfying atomic assignment exists.
The extended network $\mathcal{I}_{k+1}$ is path-consistent: every triple involving e is consistent by the atomic assignment, and every triple not involving e is consistent by the inductive hypothesis. □

Remark 5.3 (Why triple-types are insufficient)**.** If we replace (Q4) with the weaker condition that only requires the existence of *some* valid triple-type for each pair of arms (without requiring star path-consistency for *every* value), the one-point extension fails. Computational verification exhibits 43,287 counterexamples among 6,799,474 configurations on 3-arm star networks (0.64% failure rate). With the full (Q4), there are zero failures among 3,567,816 configurations.

5.3 Model existence

Theorem 5.4 (Completeness)**.** *If a quasimodel for C_0 exists, then C_0 is satisfiable.*

Proof. Construct the Henkin tree (Definition 5.1). At each expansion step, Lemma 5.2 provides consistent cross-edges. By induction, the partial model $\mathcal{I}_k$ has a path-consistent atomic RCC5 network at every finite stage.

The infinite model $\mathcal{I} = \bigcup_k \mathcal{I}_k$ satisfies:

- **Existential demands:** every $\exists R.D$ in every type is witnessed by a child node (by construction).
- **Universal constraints:** every cross-edge $\rho(e, x)$ is in $\mathsf{Safe}(\mathsf{tp}(e), \mathsf{tp}(x))$ (by domain definition).
- **Composition consistency:** every finite subnetwork is path-consistent (by Lemma 5.2). By compactness (every triple involves finitely many nodes), the infinite network is composition-consistent.

Hence $\mathcal{I} \models C_0$. □

6 Decision Procedure

Theorem 6.1 (Decidability)**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable.*

Proof. By Theorems 4.1 and 5.4, C_0 is satisfiable iff a quasimodel for C_0 exists. We decide quasimodel existence by *type elimination*:

1. Compute $\mathsf{cl}(C_0)$ and enumerate all types $T \subseteq 2^{\mathsf{cl}(C_0)}$ (at most $2^{|\mathsf{cl}(C_0)|}$).
2. Enumerate all valid pair-types, triple-types, and quadruple-types.
3. Eliminate a type τ if (Q2) fails: some existential demand has no witness type.
4. Eliminate a pair-type if (Q3) fails: some triple cannot be completed.
5. Eliminate a quadruple-type candidate if (Q4) fails.
6. Repeat steps 3–5 until stable.
7. Accept iff some surviving type contains C_0.

This procedure terminates (finitely many entities, each step eliminates at least one) and is correct (the fixed point is the maximal quasimodel). □

Theorem 6.2 (ExpTime upper bound)**.** *The type elimination procedure runs in ExpTime.*

Proof. Let $n = |\mathsf{cl}(C_0)| = O(|C_0|)$.

- Number of types: $|T| \leq 2^n$.
- Number of pair-types: $|T|^2 \times 4 = O(2^{2n})$.
- Number of triple-types: $|T|^3 \times 4^3 = O(2^{3n})$.
- Number of quadruple-types: $|T|^4 \times 4^6 = O(2^{4n+12})$.

All quantities are at most $2^{p(n)}$ for a fixed polynomial p. The elimination loop iterates at most $O(2^{4n})$ times (each iteration removes at least one entity). Each step checks a bounded number of conditions per entity. The total running time is polynomial in $2^{p(n)}$, hence $2^{q(n)}$ for some polynomial q. This is ExpTime. □

Corollary 6.3. *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is PSpace-hard (Wessel [2]) and in ExpTime.*

7 Computational Verification

All algebraic properties underlying the proof are verified by exhaustive case analysis over the RCC5 composition table. Since $|N_R^-| = 4$, the composition table has $4^2 = 16$ entries, and the space of configurations is finite.

7.1 Property 1: Pure quadruple consistency

Statement: For all $R, S_{px}, S_{xy}, S_{py}, r_{zx}$ with $r_{zx} \in \mathsf{comp}(R, S_{px})$ and $S_{py} \in \mathsf{comp}(S_{px}, S_{xy})$:

$$\mathsf{comp}(r_{zx}, S_{xy}) \cap \mathsf{comp}(R, S_{py}) \neq \emptyset.$$

Verification: 427 configurations, all pass. This is a finite proof by cases.

7.2 Property 2: Compositional monotonicity

Statement: For all R, S_{px}, S_{xy} and $S_{py} \in \mathsf{comp}(S_{px}, S_{xy})$:

$$\mathsf{comp}(R, S_{py}) \subseteq \bigcup_{s \in \mathsf{comp}(R, S_{px})} \mathsf{comp}(s, S_{xy}).$$

Verification: 164 configurations, all pass. This means every relation achievable through the direct path $p \to y$ is also achievable through $p \to x \to y$.

7.3 Property 3: Star arc-consistency

Statement: Under quadruple-type conditions (Q4), enforcing arc-consistency on the star disjunctive network never empties a domain.

Verification:

Star size	Condition	Tests	Failures
2 arms	Triple (Q3)	25,013	0
3 arms	Triple (Q3)	6,799,474	43,287
3 arms	Quadruple (Q4)	3,567,816	**0**

The triple-type condition is sufficient for 2-arm stars but fails for 3-arm stars. The quadruple condition eliminates all failures.

8 Discussion

8.1 Why quadruple-types are necessary

The standard quasimodel approach for $\mathcal{ALCI}$ uses only pair-types (or equivalently, triple-types degenerate because the model is a tree). For $\mathcal{ALCI}_{RCC5}$, the model is *not* tree-structured: all pairs of domain elements have RCC5 relations, including cross-branch pairs.

The cross-branch edges create a constraint network that must be globally consistent. Triple-types capture *pairwise* consistency: for each pair of cross-edges from a new node, there exist individually compatible values. But they do not capture *joint* consistency: a single value for one cross-edge must simultaneously have compatible partners in *all* other cross-edges.

The quadruple-type condition captures this joint consistency by requiring, for every specific value r chosen for one cross-edge, the existence of a compatible value for every other cross-edge. This is exactly the star path-consistency condition.

8.2 Relation to previous approaches

The quasimodel approach in [4] used triple-types and left the Henkin construction's cross-edge assignment as a gap. The gap is now closed by strengthening (Q3) to (Q4).

The tableau approaches [5, 6] extract quasimodels from open branches. Their soundness gap (model extraction) is resolved by the same mechanism: the quadruple-type condition in the quasimodel ensures model existence.

8.3 Extension to $\mathcal{ALCI}_{RCC8}$

RCC8 also has the patchwork property (for path-consistent atomic networks) and a tractable satisfiability check via path-consistency (Renz & Nebel 1999). The quadruple-type approach extends directly: replace the RCC5 composition table with the RCC8 table, and verify the analogous algebraic properties (Properties 1–3). We expect the same result but leave the verification to future work.

9 Erratum: Soundness Gap in (Q4)

9.1 The gap

The proof of Theorem 4.1 (Q4) contains an error. The proof states: "choose witnesses z, p, x_i, x_j in the model with these types." This provides four elements with the types $\tau_z, \tau_p, \tau_i, \tau_j$. However,

Q4 also specifies *particular relations* $R \in \mathsf{Safe}(\tau_z, \tau_p)$, $R_{pi} \in \mathsf{Safe}(\tau_p, \tau_i)$, etc., and the proof requires these to be the *actual* pairwise relations between the chosen elements.

In a complete-graph model, the relation between any two elements is determined by the model, not chosen by the prover. Given elements z, p with $\mathsf{tp}(z) = \tau_z$, $\mathsf{tp}(p) = \tau_p$, the actual relation $\rho(z, p)$ is some fixed $R' \in \mathsf{Safe}(\tau_z, \tau_p)$, which may or may not equal the R specified in Q4. The proof implicitly assumes that all safe relation combinations are realized in the model, which is not guaranteed.

9.2 Computational confirmation

A working implementation (`alcircc5_reasoner.py`) confirms the gap. The concept

$$C_{15} \;=\; \exists\mathsf{PP}.(\forall\mathsf{DR}.A) \;\sqcap\; \exists\mathsf{DR}.\neg A$$

is satisfiable (the PP-witness with $\forall\mathsf{DR}.A$ and the DR-witness with $\neg A$ can be PO-connected, avoiding the $\forall\mathsf{DR}$ constraint). However, Q4 as stated rejects all type sets containing these types: the universal quantification over all safe relations produces configurations where no consistent r exists, even though the model only uses a subset of safe relations.

9.3 The disjunctive path-consistency alternative

Replacing Q4 with *disjunctive path-consistency* of the Safe constraint network over the type set fixes the soundness direction: extracting types from a model always yields a path-consistent network (the model's atomic relations provide a satisfying assignment that refines the Safe domains). By the patchwork property, path-consistent disjunctive RCC5 networks are satisfiable, so this check is both sound and complete at the type level.

The implementation using this approach correctly decides all 18 test concepts, including C_{15}.

9.4 What survives and what does not

Result	**Status**	**Comment**
Theorem 4.1 (Q1–Q3)	Correct	Standard
Theorem 4.1 (Q4)	**Gap**	Unrealized relations
Lemma 5.2	Correct *if* Q4 holds	Completeness direction
Theorem 5.4	Correct *if* Q4 holds	Follows from lemma
Theorem 6.1	**Strong evidence**	See below
Properties 1–3 (Section 7)	Correct	Pure algebraic facts
Three-check algorithm	**New**	713 concepts, 0 errors

The completeness direction (quasimodel $\Rightarrow$ model via the one-point extension lemma) is correct *conditional on* Q4 holding. The gap is exclusively in the soundness direction: a model does not necessarily yield a type set satisfying Q4.

The disjunctive path-consistency check restores soundness but is not sufficient on its own for completeness: the one-point extension lemma requires that when adding a new witness e as an R-child of d, the *star network* $\{D(e, x) : x \in \text{existing}\}$ where $D(e, x) = \mathsf{comp}(\mathsf{inv}(R), \rho(d, x)) \cap \mathsf{Safe}(\mathsf{tp}(e), \mathsf{tp}(x))$ is path-consistent. Greedy witness selection can produce empty star domains, even when alternative witness types succeed.

9.5 The three-check algorithm

The working implementation replaces Q4 with three *necessary* conditions on the quasimodel type set T. Each condition is extractable from any model of the concept, so UNSAT answers are **provably sound**.

Check 1: Disjunctive path-consistency. The Safe constraint network over T must be path-consistent: for each triple of types (τ_a, τ_b, τ_c), every relation $r \in \mathsf{Safe}(\tau_a, \tau_b)$ must be supported through τ_c, i.e., $r \in \mathsf{comp}(r_1, r_2)$ for some $r_1 \in \mathsf{Safe}(\tau_a, \tau_c)$, $r_2 \in \mathsf{Safe}(\tau_c, \tau_b)$. By the patchwork property (Renz & Nebel 1999), this is both sound and complete at the type level.

Check 2: Sibling compatibility. For each type τ_i with demands $(R_1, D_1), \ldots, (R_k, D_k)$, the type set must contain witness types $\tau_{j_1}, \ldots, \tau_{j_k}$ such that *all pairs* satisfy:

$$\mathsf{comp}(\mathsf{inv}(R_m), R_{m'}) \cap \mathsf{Safe}(\tau_{j_m}, \tau_{j_{m'}}) \neq \emptyset \qquad \text{for all } m \neq m'.$$

This ensures that siblings (children of the same parent) can be assigned a consistent mutual relation. Crucially, when two demand slots use different roles ($R_m \neq R_{m'}$), the witnesses must be *distinct elements* even if they share the same type; the check must verify the pairwise constraint in all such cases (same-type bypass is valid only when $R_m = R_{m'}$, allowing a single element to serve both demands). The check uses arc-consistency pruning followed by backtracking search.

Check 3: Role-path compatibility. For each chain $g \xrightarrow{R} j \xrightarrow{S} w$ in the quasimodel tree (grandparent–parent–child), the grandchild w must be connectable to the grandparent g:

$$\mathsf{comp}(\mathsf{inv}(S), \mathsf{inv}(R)) \cap \mathsf{Safe}(\tau_w, \tau_g) \neq \emptyset.$$

The check iteratively prunes invalid witnesses until a fixpoint. This catches cross-level constraints such as $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ forcing a relation outside Safe.

9.6 Computational verification

A *witness plan* precomputes, for each type, the sibling-compatible witness assignment via backtracking CSP. The Henkin tree builder uses this plan with *future-flexibility ordering* (preferring PO over DR over PP/PPI for cross-relations, since PO gives the widest average composition) and backtracking over alternative witnesses.

Result. A comprehensive stress test of **713 concepts** across seven categories (known SAT, known UNSAT, adversarial cross-role universals, systematic role–atom combinations, and random concepts of depths 2–4) finds:

- **Zero correctness errors**: all known-SAT and known-UNSAT concepts are classified correctly.
- **Zero Henkin tree failures** for all 16 basic test concepts across **12 independent hash seeds**.
- 7 remaining Henkin failures in the stress test occur at depths 2–3 for concepts that *are* satisfiable (verified by explicit cyclic models) but whose tree unfoldings have cross-level composition conflicts. One additional failure is a builder over-branching issue (the concept has a tree model). None are reasoner soundness bugs.

Concepts requiring cyclic models. Seven concepts are satisfiable only in cyclic (non-tree) models. All share the structure $\exists R.\varphi \sqcap \exists \mathsf{PP}.\psi \sqcap \forall S.\chi$ where the leaf type τ_ℓ (satisfying ψ) has $\mathsf{Safe}(\tau_\ell, \tau_\ell) = \{\mathsf{DR}\}$ due to accumulated universal constraints, while $\mathsf{comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{PPI}, \mathsf{PP}, \mathsf{PO}\}$ forces non-DR cross-level relations between leaf copies. Since $\{\mathsf{DR}\} \cap \{\mathsf{PPI}, \mathsf{PP}, \mathsf{PO}\} = \emptyset$, no tree model exists. All seven have verified 3-element cyclic models (e.g., $d\ \mathsf{PO}\ e$, $d\ \mathsf{PP}\ f$, $e\ \mathsf{PP}\ f$):

$$\begin{aligned}
&\exists \mathsf{DR}.A \sqcap \exists \mathsf{PP}.\neg A \sqcap \forall \mathsf{PO}.A \\
&\exists \mathsf{PO}.A \sqcap \exists \mathsf{PP}.\neg A \sqcap \forall \mathsf{PO}.A \\
&\exists \mathsf{PO}.A \sqcap \exists \mathsf{PP}.\neg B \sqcap \forall R.A \qquad (R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\};\ \text{4 concepts}) \\
&\exists \mathsf{PP}.A \sqcap \exists \mathsf{PP}.\neg A \sqcap \forall \mathsf{PO}.A
\end{aligned}$$

9.7 Theoretical status

What is proved:

- All three checks are *necessary* conditions (extractable from any model).
- UNSAT answers are provably sound.
- The one-point extension lemma is correct conditional on its star path-consistency hypothesis.
- The algebraic properties are unconditionally correct.

What is conjectured:

- The three checks are jointly *sufficient* for the one-point extension at every tree depth.
- Specifically: sibling compatibility ensures non-empty star domains for siblings; role-path compatibility ensures non-empty star domains for grandparents; and the patchwork property ensures the resulting disjunctive star network is globally satisfiable.

What remains: A formal proof that the three checks imply star path-consistency for the full star network at each extension step (including uncle and cousin constraints). Additionally, the stress test reveals that some satisfiable concepts may require *cyclic models* rather than tree models: when $\mathsf{Safe}(\tau, \tau)$ is narrow (e.g., $\{\mathsf{DR}\}$ only) and the tree-path composition forces a different relation, tree unfolding fails even though a small cyclic model exists. The soundness proof should account for this by using the patchwork property to construct finite (possibly cyclic) models rather than relying on tree-shaped Henkin constructions exclusively.

10 Conclusion

The quadruple-type elimination approach to $\mathcal{ALCI}_{RCC5}$ satisfiability introduces the right structural idea — star path-consistency captures the interaction that triple-types miss — but condition (Q4) as formulated is too strong: models do not satisfy it (Section 9). Replacing Q4 with three necessary conditions — *disjunctive path-consistency*, *sibling compatibility*, and *role-path compatibility* — fixes soundness (provably: all three conditions are extractable from any model) and provides strong computational evidence for completeness.

A working implementation correctly decides all test concepts in a comprehensive stress test of 713 concepts. The sibling compatibility check correctly identifies cross-role universal contradictions (e.g., $\exists\mathsf{PP}.(\forall\mathsf{PPI}.A) \sqcap \exists\mathsf{PPI}.\neg A$ is UNSAT because $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ forces the universal to fire). The Henkin tree extension test passes with **zero failures** for all basic tests across 12 hash seeds.

Decidability status. The algorithm is a *candidate decision procedure* for $\mathcal{ALCI}_{RCC5}$ with the following properties:

1. UNSAT answers are **provably sound** (all checks are necessary conditions).
2. SAT answers are supported by **strong computational evidence** (zero counterexamples across extensive testing) but lack a formal sufficiency proof.
3. The algorithm runs in ExpTime (exponential in closure size, polynomial in the number of types).

The decidability of $\mathcal{ALCI}_{RCC5}$ is therefore strongly *conjectured* but not formally *proved*. The remaining gap is a formal proof that the three checks jointly imply star path-consistency at every extension step of the Henkin construction.

All algebraic properties are verified by exhaustive case analysis over the RCC5 composition table (427 + 164 + 3,567,816 configurations, all zero failures under the quadruple condition). This constitutes a finite proof by cases for the base algebraic facts, and exhaustive verification for the arc-consistency stability.

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